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JAVA PROJECT REPORT ON

Hospital Bed Allocation and Availability Monitoring System

Submitted by

YUKESH.M

25BCS162

Under the Guidance of

Mr.ARAVIND

Technical Trainer

Technical Department, KG College of Arts and Science

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DECLARATION

I hereby declare that the project entitled " Hospital Bed Allocation and Availability Monitoring System " submitted in partial fulfillment of the requirements for the course **Java programming** is a **record of my original work** carried out under the guidance of Mr.**ARAVIND** , Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

YUKESH.M

CERTIFICATE

This is to certify that the project entitled **“ Hospital Bed Allocation and Availability Monitoring System ”** is a **bona fide work** carried out by **YUKESH 1-B**, **Department of Computer Science**, , **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming**..

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: SARAVANAMPATTI

Date:10/04/2026

Submitted for the Viva-Voce Examination held on _____10/04/2026____

Guide:	**Head of the Department**
(Technical Trainer, KGCAS)	(Technical Head, KGCAS)
Mr. ARAVIND	Mr. BOOPALASUNDHARAMOORTHI

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ABSTRACT

The Hospital Bed Allocation and Availability Monitoring System is a Java-based application designed to manage hospital bed allocation efficiently. In many hospitals, bed management is done manually, which can lead to errors, delays, and inefficient utilization of resources.

This project aims to provide an automated system that tracks bed availability and allows quick allocation and release of beds. The system maintains details such as bed ID, occupancy status, and patient information.

The application is developed using Core Java concepts such as classes, objects, methods, loops, and conditional statements. It provides a menu-driven interface where users can view bed status, allocate beds, and release beds.

The main objective of the system is to improve efficiency, reduce manual errors, and provide real-time monitoring of hospital beds..

This project demonstrates the practical implementation of **object-oriented programming concepts in Java** and provides a basic foundation for developing more advanced restaurant management systems in the future.

1. INTRODUCTION

In hospitals, proper management of bed allocation is essential for providing timely treatment to patients. Many hospitals still use manual methods to track bed availability, which can lead to confusion, delays, and errors.

With the advancement of technology, automated systems have become necessary to manage hospital resources efficiently. A computerized system helps hospital staff quickly check bed availability and assign beds to patients without errors.

The Hospital Bed Allocation System is designed to automate this process. It allows users to manage beds efficiently by tracking availability, allocating beds, and releasing them when patients are discharged.

The system is developed using Java programming and demonstrates the application of object-oriented programming in real-world scenarios.

This project is developed using **Core Java programming concepts** such as classes, objects, methods, conditional statements, loops, and exception handling. The application can be implemented either as a **console-based program or using Java Swing for a graphical user interface (GUI)**.

The primary goal of this system is to reduce manual effort, improve billing accuracy, and provide faster service to customers. This project also demonstrates how Java programming can be applied to develop simple real-world business applications.

1.1 Objective

- The primary objective of this project is to design and develop a reliable and efficient system for managing hospital bed allocation using Java programming. The system aims to provide real-time information about bed availability, allowing hospital staff to quickly allocate beds to patients and release them when no longer needed. Another important objective is to reduce the chances of human error that commonly occur in manual systems. The project also focuses on improving workflow within hospitals by providing a simple and user-friendly interface. Additionally, it aims to demonstrate the practical implementation of object-oriented programming concepts such as classes, objects, and methods in solving real-world problems. Overall, the system is intended to enhance efficiency, accuracy, and decision-making in hospital management .

1.2 Overview

The Hospital Bed Allocation System is a console-based application developed using Java programming language. The system is designed to maintain records of hospital beds, including their availability status and assigned patient details. It provides a menu-driven

interface where users can perform operations such as viewing available beds, allocating beds to patients, and releasing beds when patients are discharged. The system uses arrays or objects to store bed details and applies logical operations to manage them efficiently. The overall workflow of the system is simple: the user selects an option from the menu, the system processes the request, and the result is displayed on the screen. This project provides a basic yet effective solution for managing hospital beds and can be further expanded into a complete hospital management system.

1.3 Features

The system includes several useful features that improve hospital bed management. It provides real-time tracking of bed availability, allowing staff to quickly identify free and occupied beds. The system supports easy allocation of beds to patients by entering necessary details such as patient name and bed ID. It also allows releasing beds when patients are discharged, automatically updating the status. Another important feature is the display of all bed details, which helps in monitoring overall usage. The system is designed to be simple and user-friendly, making it easy to operate. Additionally, it reduces paperwork and manual effort, improving accuracy and efficiency in hospital operations.

2.PROGRAM ANALYSIS

Program analysis is the process of understanding the problem, identifying requirements, and determining the best approach to solve it. In this project, the need for an automated hospital bed management system is analyzed. The existing manual system is studied to identify its limitations, such as inefficiency, lack of real-time updates, and high chances of errors. Based on this analysis, a software solution is proposed to address these issues. The analysis also includes understanding user requirements, such as ease of use, quick access to information, and accurate data handling. This phase ensures that the developed system meets the needs of users and performs efficiently Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in the existing system and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

The main problem addressed in this project is the inefficient management of hospital beds using manual methods. Hospitals often face challenges in tracking bed availability, especially during peak hours or emergencies. Manual systems are prone to errors such as incorrect entries, duplication, and delays in updating information. This can lead to confusion and inconvenience for both staff and patients. The lack of a centralized system makes it difficult to monitor bed usage effectively. Therefore, there is a need for an automated system that can manage bed allocation accurately and provide real-time updates.

2.2 Project Overview

This project provides a simple solution to manage hospital bed allocation using Java programming. The system maintains a list of beds and their current status, whether available or occupied. Users can interact with the system through a menu-driven interface to perform various operations. The system ensures that no two patients are assigned the same bed and that all updates are reflected immediately. It is designed to be efficient, reliable, and easy to use. This project serves as a basic model that can be expanded into a more advanced system with additional features.

2.3 Scopes and Limitations

Scope

The scope of this project includes managing hospital beds efficiently within a small or medium-sized hospital. It provides basic functionalities such as allocation, release, and monitoring of beds. However, the system has some limitations. It is a console-based application and does not include a graphical user interface. The data is stored temporarily and

is not saved permanently in a database. The system also does not handle advanced features such as patient records, billing, or staff management. Despite these limitations, the project provides a strong foundation for developing more advanced systems in the future.

3.SYSTEM DESIGN

System design defines the structure and organization of the application. In this project, the system is designed using a modular approach to improve clarity and maintainability. Each module performs a specific function, such as input handling, processing, and output display. The design includes a class to represent hospital beds, with attributes such as bed ID, status, and patient name. The system uses a menu-driven interface for user interaction. This structured design ensures that the system is easy to understand, implement, and modify.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

The main modules of the Hospital Bed Allocation and Availability Monitoring System are explained below:

- **Input Module**

The Input Module is responsible for collecting all the necessary data from the user. This includes details such as patient name, bed number, and the user's choice from the menu options. The module ensures that the input provided by the user is valid and meaningful before passing it to the processing module. For example, it checks whether the entered bed number exists or not and ensures that invalid inputs are not accepted. This module plays a crucial role in maintaining the accuracy of the system, as incorrect input can lead to wrong results. It acts as the starting point of the system and ensures smooth interaction between the user and the application.

- **Processing Module**

The Processing Module is considered the core of the system, as it handles all the logical operations. This module processes the input data and performs actions such as checking bed availability, allocating a bed to a patient, and releasing a bed when it becomes free. It uses conditions and control statements to ensure that operations are performed correctly. For example, before allocating a bed, the system checks whether the bed is already occupied or available. If the bed is free, it assigns it to the patient; otherwise, it displays an error message. This module ensures that all operations are performed accurately and efficiently.

- **Output Module**

The Output Module is responsible for displaying the results of the system to the user. It shows information such as available beds, occupied beds, patient details, and confirmation messages for operations like allocation and release. The output is displayed in a clear and understandable format so that users can easily interpret the results. This module improves user experience by providing proper feedback for every action performed in the system. For example, after allocating a bed, it displays a success message along with the updated status of the bed.

- **File Handling Module**

The File Handling Module is used to store and retrieve data permanently. In the basic version of the project, data is stored temporarily in memory and is lost when the program is closed. However, with file handling, the system can save details such as bed status and patient information in a file. This allows the system to retain data even after restarting the application. It improves the reliability and usability of the system in real-world scenarios. This module uses file operations such as reading and writing data to maintain records efficiently.

Overall Working of Modules

All these modules are interconnected and work together to perform the system operations. First, the Input Module collects data from the user. Then, the Processing Module analyzes and processes the data. After processing, the Output Module displays the result. .

4.METHODOLOGY

The methodology describes the step-by-step process followed to develop the system. It begins with problem analysis, followed by system design and implementation. The program is developed using Java programming language, applying object-oriented principles. After development, the system is tested to ensure correctness and reliability. Finally, the results are documented, and the system is evaluated. This structured approach ensures successful project completion.

The development process includes the following stages:

• Problem Analysis
• System Design
• Program Development
• Testing and Validation
• Result and Documentation

4.1. Problem Analysis

Problem analysis is an essential step in identifying the issues present in the current system and determining the requirements for the new system. In this project, the existing manual method of managing hospital beds is analyzed in detail. It is observed that manual systems are time-consuming, prone to human errors, and lack real-time updates. During emergencies, it becomes difficult for hospital staff to quickly identify available beds, leading to delays in patient care. These problems highlight the need for an automated system that can manage bed allocation efficiently. The analysis helps in understanding user requirements such as accuracy, speed, and ease of use, which are essential for designing an effective solution.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

• Bed Display Module
• Allocate bed Module
• View bed Module

Bed Release Module
• Exit Module

The program flow and structure are designed so that users can easily interact with the system.

4.3. Program Development

After designing the system, the program is implemented using the **Java programming language**.

The following steps were followed during development:

• Creation of classes and methods
• Implementation of menu-driven logic
• Writing code for bill calculation
• Handling user input
• Displaying the final bill

Core Java concepts such as **classes, objects, loops, conditional statements, and methods** were used during development.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different types of testing methods were applied in this project:

- **Unit Testing:** Each module (input, processing, output) was tested separately to ensure it works correctly.
- **Integration Testing:** All modules were tested together to verify proper interaction between them.
- **System Testing:** The entire system was tested as a whole to ensure it meets the project requirements.
- **User Acceptance Testing:** The system was tested from the user's perspective to ensure it is easy to use and understand.

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

The results obtained from the system are carefully analyzed and documented in this phase. The system successfully performs all operations such as displaying bed availability, allocating beds to patients, and releasing beds when required. The output is displayed clearly on the console, making it easy for users to understand. Documentation includes detailed explanations of the system design, implementation, and results. It also includes screenshots of input and output, which help in understanding the working of the system. Proper documentation ensures that the project can be easily understood and evaluated.

4.6. Algorithm

Algorithm for the **Hotel & Restaurant Billing System**:

1. Start

- Initialize beds
- Display menu
- Get user input
- Perform operation
- Repeat
- Stop

4.7. Flowchart

Students should draw a flowchart showing program flow.

Example flow:

Start
↓
Display Menu
↓
input
↓Process

↓
Display output
↓
End

5.SOURCE CODE

```
package viva;
       import java.util.Scanner;
       class Bed {
           int bedId;
           boolean isOccupied;
String patientName;
Bed(int id) {
 this.bedId = id;
this.isOccupied = false;
 this.patientName = "";
           }
 void allocate(String name) {
this.isOccupied = true;
 this.patientName = name;
           }
void release() {
this.isOccupied = false;
this.patientName = "";
           }
void display() {
if (isOccupied) {
 System.out.println("Bed ID: " + bedId + " | Occupied by: “+ patientName);
} else {
 System.out.println("Bed ID: " + bedId + " | Available");
               }
           }
       }
public class HospitalSystem {
           public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
```

```
System.out.print("Enter number of beds: ");
int n = sc.nextInt();
 Bed[] beds = new Bed[n];
for (int i = 0; i < n; i++) {
beds[i] = new Bed(i + 1);
 }

int choice;
do {
                  System.out.println("\n--- Hospital Bed System ---");
                   System.out.println("1. View Beds");
                   System.out.println("2. Allocate Bed");
                   System.out.println("3. Release Bed");
                   System.out.println("4. Exit");
                   System.out.print("Enter choice: ");
choice = sc.nextInt();
  switch (choice) {
                      case 1:
                          for (Bed b : beds) {
                              b.display();
                          }
                          break;
                      case 2:
                          System.out.print("Enter bed ID to allocate: ");
                          int id = sc.nextInt();
                          sc.nextLine(); // consume newline
.                                   14
                          if (id > 0 && id <= n) {
                              if (!beds[id - 1].isOccupied) {
                                  System.out.print("Enter patient name: ");
                                  String name = sc.nextLine();
                                  beds[id - 1].allocate(name);
                              System.out.println("Bed allocated successfully.");
```

```
                } else {
                    System.out.println("Bed already occupied.");
                }
            } else {
                System.out.println("Invalid Bed ID.");
            }
            break;

        case 3:
            System.out.print("Enter bed ID to release: ");
            int rid = sc.nextInt();

            if (rid > 0 && rid <= n) {
                if (beds[rid - 1].isOccupied) {
                    beds[rid - 1].release();
                System.out.println("Bed released successfully.");
                } else {
                    System.out.println("Bed is already free.");
                }
            } else {
                System.out.println("Invalid Bed ID.");
            }
            break;

        case 4:
            System.out.println("Exiting...");
```



```
            break;

        default:
            System.out.println("Invalid choice.");
    }
```

```
        } while (choice != 4);


        sc.close();
    }
}
```

6.INPUT AND OUTPUT SCREENS

INPUT SCREEN:

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

D:\JavaPrograms\prg11>javac HospitalSystem.java

D:\JavaPrograms\prg11>java HospitalSystem
Enter number of beds: 10

--- Hospital Bed System ---
1. View Beds
2. Allocate Bed
3. Release Bed
4. Exit
Enter choice: 2
```

OUTPUT SCREEN:

.

```
C:\Windows\System32\cmd.e

Bed ID: 1 | Occupied by: ram
Bed ID: 2 | Available
Bed ID: 3 | Available
Bed ID: 4 | Available
Bed ID: 5 | Available
Bed ID: 6 | Available
Bed ID: 7 | Available
Bed ID: 8 | Available
Bed ID: 9 | Available
Bed ID: 10 | Available

--- Hospital Bed System ---
1. View Beds
2. Allocate Bed
3. Release Bed
4. Exit
Enter choice: 3
Enter bed ID to release: 1
Bed released successfully.

--- Hospital Bed System ---
1. View Beds
2. Allocate Bed
3. Release Bed
4. Exit
Enter choice: 1
Bed ID: 1 | Available
Bed ID: 2 | Available
Bed ID: 3 | Available
Bed ID: 4 | Available
Bed ID: 5 | Available
Bed ID: 6 | Available
Bed ID: 7 | Available
Bed ID: 8 | Available
Bed ID: 9 | Available
Bed ID: 10 | Available

--- Hospital Bed System ---
1. View Beds
2. Allocate Bed
3. Release Bed
4. Exit
Enter choice: 4
Exiting...
```

7.DISCUSSION & CONCLUSION

7.1 Discussion

The Hospital Bed Allocation and Availability Monitoring System was developed and tested to evaluate its performance in managing hospital bed resources efficiently. The system successfully performs all key operations such as displaying bed availability, allocating beds to patients, and releasing beds when they become free. During testing, it was observed that the system provides accurate and real-time updates, which helps in reducing confusion and delays that are commonly seen in manual systems. The menu-driven interface makes the system easy to use, even for users with minimal technical knowledge

7.2 Conclusion

The Hospital Bed Allocation and Availability Monitoring System is a successful implementation of a software solution designed to improve hospital resource management. The project achieves its primary objective of providing an efficient and reliable system for tracking and managing hospital beds. By automating the process of bed allocation and
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monitoring, the system reduces manual workload, improves accuracy, and ensures better utilization of available resources.

The use of Java programming and object-oriented concepts has made the system structured, organized, and easy to understand. The modular design allows each component of the system to function independently, making it easier to maintain and enhance in the future. The system demonstrates how simple programming techniques can be applied to solve real-world problems effectively.

8.REFERENCES

The following resources were used during the development of this project:

Books

1. Java: The Complete Reference Author: Herbert Schildt

   Publisher: McGraw-Hill Education

2. Head First Java Authors: Kathy Sierra & Bert Bates

   Publisher: O'Reilly Medi

Website

3. Oracle Java Documentation Website: https://docs.oracle.com/javase/

4. GeeksforGeeks Website: https://www.geeksforgeeks.org/

5. W3Schools Java Tutorial Website: https://www.w3schools.com/java/

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